

VSLM: Virtual Sound-Level Meter

Want to play around with the acoustic signals or the sound inputs for your audio device? This article guides you on how to start working with VSLM, a tool that allows you to do just that

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Sound-level meters are expensive instruments that measure sound or acoustic pressure, and are used to analyse sound signals. Thankfully, there is now a software tool available that allows students to analyse acoustic signals without the need for expensive full-featured sound level meters.

VSLM (Virtual Sound Level Meter) is a program that provides students, educators, researchers and consultants with a tool for analysing calibrated sound level recordings. A user can make calibrated digital recordings of an acoustic signal, and analyse using the software with multiple analysis techniques that allow students to 'play' with the analyser and observe what the different settings do to the signals.

VSLM follows the ANSI acoustic measurement and analysis standards, which ensure that the results obtained using VSLM match the results obtained from any other standard measurement equipment. The current version of VSLM implements basic integrating, 1/3 octave band/high-resolution FFT sound level meter functions, noise dose analysis and spectrograms.

The GUI

The software's GUI is made to resemble it with a physical sound analyser. Fig. 1 shows the main GUI. It consists of a main menu (1) at the top for saving and loading various settings. There is a set of analysis buttons (2), which

System requirements

OS: Windows XP and above

MATLAB requirements (if you want to modify the source code): MATLAB R12 or later (with signal processing toolbox)

RAM: 1 GB

Note: The installation file can be found in the EFY Plus DVD that accompanies this magazine.

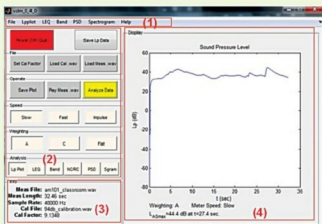


Fig. 1: The main GUI

is used for various analysis modes. The info box (3) provides the details of currently selected measurements and calibration. Analysed plots and computed information are displayed in the display area (4).

Calibration

The input .wav files should be calibrated for the analysis. .wav files are generally 16- or 24-bit integers and their inputs need to be converted into Pascals by setting a calibration constant/factor; while for the files that are already calibrated (like some floating point .wav files), the calibration factor should be set to 1.

the calibration is then calculated by the system, and the calibration factor is set accordingly. The calibration file name and calibration factor are displayed in the info box (3).

Calibration files are recordings from sound-level calibrators with sound levels of 94, 114 or 124 dB, and the loaded calibrations should be in .wav file format at sample rates of 22.05 kHz, 44.1 kHz, 48 kHz or 96 kHz.

Set Cal factor. If you can estimate the calibration factor for any recording, use this button to manually set the calibration factor. This is particularly useful while performing re-analysis of recordings whose calibration factor

Note: VSLM's default calibration factor is 1.

Load Cal .wav.

The Load Cal .wav button can be used to load a .wav file that has a calibration recording. After pressing the button, you need to select the calibration recording and enter the calibration level in dB. The rms value of